Status	Finished
Started	Sunday, 18 October 2026, 7:56 PM
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Duration	7 mins 42 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Merge Sort
- b. Quick Sort
- c. Insertion Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It compares adjacent elements and swaps them if necessary
- c. It divides the list into two halves, sorts each half, and then merges them
- d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **3**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- b. A sorting algorithm is stable if it preserves the order of all keys
- c. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **4**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Greedy method
- b. Dynamic programming
- c. Divide and conquer
- d. Backtracking

The correct answer is:
Divide and conquer

Question **5**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Searching
- b. Rearranging
- c. Merging
- d. Sorting

The correct answer is: Sorting

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort
b. Merge Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Binary Search
b. Bubble Sort
c. Merge Sort
d. Linear Search

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Bubble Sort
b. Binary Search
c. Merge Sort
d. Linear Search

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Deleting data from a list
b. Inserting data into a list
c. Arranging data in a particular format
d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **10**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are compared element by element
c. They are combined to form a single sorted list
d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **11**

Complete

Mark 0.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It does not require recursion
b. It sorts data using a single pass
c. It is less efficient than bubble sort
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **12**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 80,150,200,470,40,90,300,400,70,120
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **13**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is less efficient for large lists
- b. Bubble Sort is difficult to implement
- c. Bubble Sort does not guarantee sorted order
- d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes code execution slower
- c. Decreases the efficiency of algorithms
- d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort

b. Merge sort

c. Selection sort

d. Quick sort

The correct answer is:
Merge sort